

SUSHI ASSEMBLY LINE

Midterm Updates



MEET THE TEAM



NICOLAS JAO

Smart Systems Automation

Previous Experience:

Engineering Co-op at Sofina Foods

Role:

Primary: Mechanical Design
Secondary: Controls



JACOB KWAN

Smart Systems Automation

Previous Experience:

Automation Service Solutions Co-op at Eclipse Automation

Role:

Primary: Electrical & Controls Design



JAMIE WILLIAMS

Industrial Systems Automation

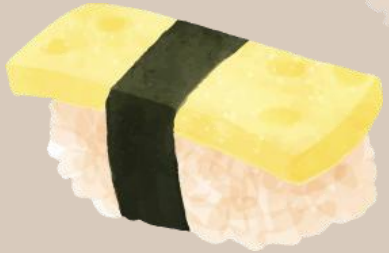
Previous Experience:

Engineering Intern at Auper Electric Motorcycles

Role:

Primary: Mechanical Design

01



OUR PROBLEM STATEMENT

THE PROBLEM

SCENARIO

Basic sushi rolls are high demand at sushi restaurants, but takes long to make

VALUE

Automating menial, repetitive tasks frees the skilled chefs' time, allowing them to focus on more complex dishes

TARGET

For use in sushi restaurants or mass-produced sushi assembly lines for grocery stores



02

OUR OBJECTIVES AND DESIGNS



OBJECTIVES

HIGH-QUALITY

Produce consistent rolls that are uniform in shape

01





02

FOOD-GRADE

Equipment used allows for safe consumption

SAFE

Humans can safely interact with the parts of the assembly line

03





04

MODULAR

Parts can be modified to allow for changes according to desired designs

5 STATIONS:

00 *Foundation & Set Up*

01 *Flattening the Rice*

02 *Placing the Neta*

03 *Rolling the Sushi*

04 *Cutting the Roll*

05 *Plating the Pieces*



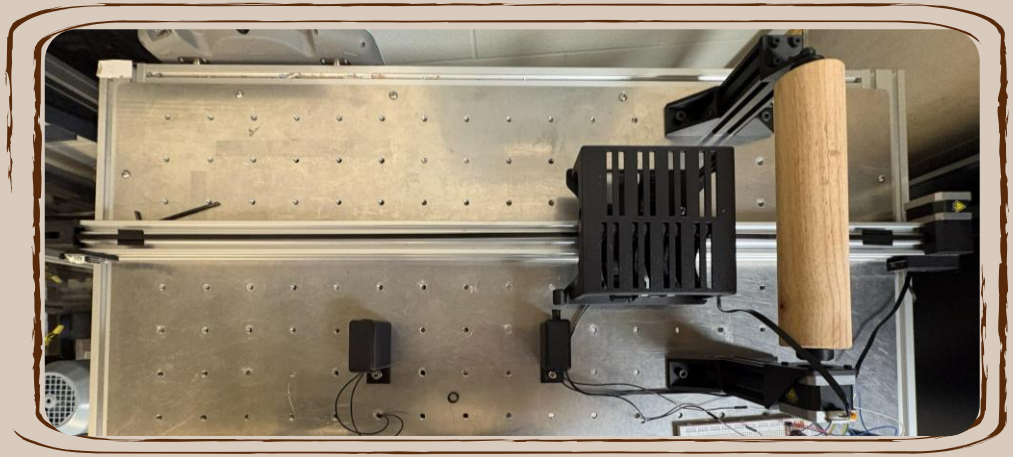
0. FOUNDATION & SET UP

Function: To carry the sushi roll from station to station across the table.

- Belt driven using a 42-34 Stepper motor
- Controlled with an A4988 stepper motor driver
- Controlled across 40x40mm V-slot 1m aluminum extrusion
- Uses limit switches to detect platform position at ends, and reed switches in between

Initial set up:

- Place nori on the platform
- Scatter rice on top of the nori
- Press Start on the HMI on the laptop



1. FLATTENING THE RICE



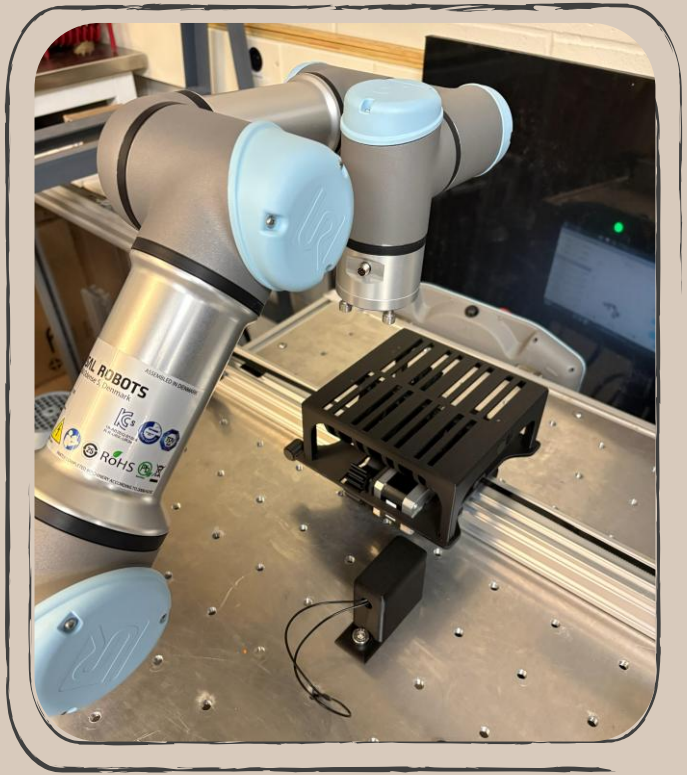
Function: Press the rice onto the nori to create a uniform sheet of rice

- Utilizes a wooden rolling pin attached to a motor
- Platform moves slowly under the rolling pin
- Station is completed as indicated by reed switch 1

2. PLACING THE NETA (TOPPINGS)

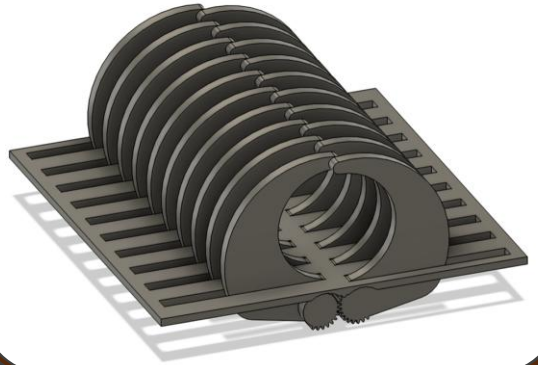
Function: Drops the provided toppings onto the sheet of rice in the preprogrammed position

- Uses the UR3e cobot to consistently pick and place the topping onto the exact same position every time
- Toppings will be prepared and placed in a bin next to the robotic arm
- Reed switch 1 sends a signal to the robotic arm to begin its program
- Signal is sent back to Arduino to continue once completed



3. ROLLING THE SUSHI





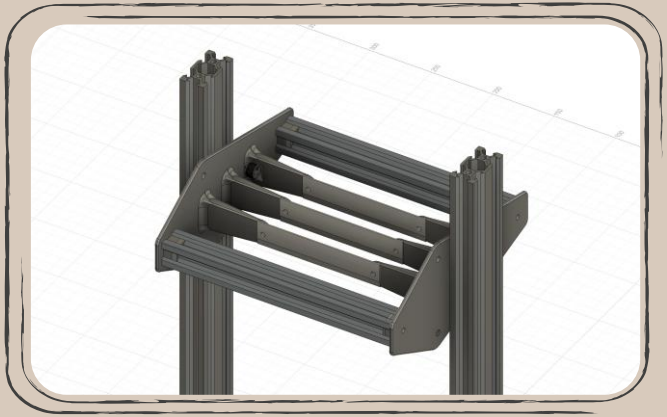
Function: Claws come from under the platform to gather the ingredients and press into a tight roll

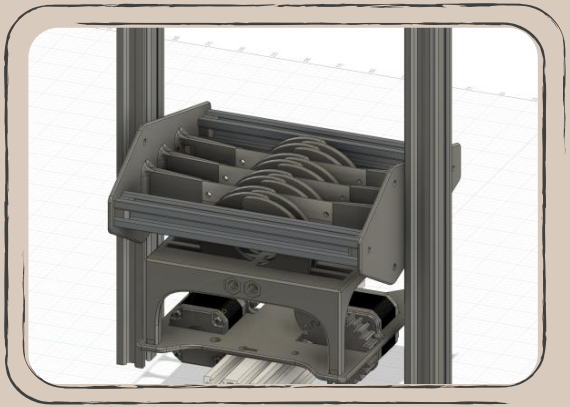
- “Ribcage” design with prongs that collapse into the roll from the bottom
- A high-torque stepper motor will control each set of claws, allowing individual control
- Mounted on linear slide platform and held up by long spacers to give room to the ribs in rest position
- Triggered when the platform reaches reed switch 2

4. CUTTING THE ROLL

Function: Set of blades come down Slice the sushi roll into 4 evenly sliced pieces

- Uses 3 evenly spaced, thin blades parallel to one another held within a frame
- Will have a plastic safety guard
- Moves straight up and down within metal frame (extrusions)
- Driven by stepper motor and lead screw to convert the rotational force into linear force

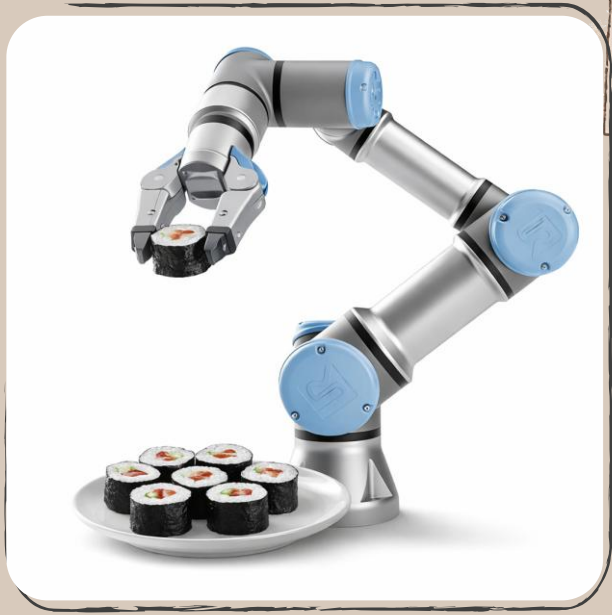




5. PLATING THE ROLLS

Function: A robot picks and places the finished sushi pieces onto a plate, ready to serve.

- Platform moves to the end limit switch and the operator is presented with the 4 finished maki sushi rolls
- Option for robotic arm to plate the rolls
- Once cleared, HMI is used to reset the platform back to the beginning and go into idle state



03

WHAT WE’VE ACHIEVED



THE TIMELINE

WEEKS 1-2

- *Recalculate scope*
- *Finish CAD for rice flattening station*
- *3D print necessary parts*

WEEK 3-4

- *Order additional motors*
- *Test rolling station with ingredients*
- *Program rolling station motors*

WEEKS 5-6

- *Install rice flattening station*
- *Start design for end-effector on robotic arm*
- *Start CAD for slicing station*

WEEK 7

- *Test final rice rolling station*
- *3D print necessary parts*
- *Midterm report and presentation*

WEEKS 8-9

- *Finish CAD for slicing station*
- *Finalize master assembly CAD*
- *Install/program slicing station*
- *Source end-effector for robotic arm*

WEEK 10-11

- *Test slicing station*
- *Install robotic arm end-effector*
- *Incorporate master control program & HMI*

WEEK 12

- *Program robotic arm*
- *Troubleshoot full system*

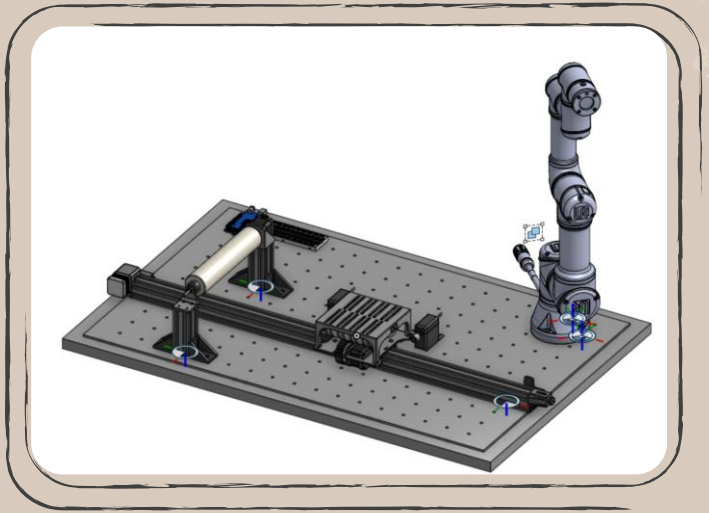
WEEK 13

- *Final testing*
- *Poster presentations*
- *Submitting final report*

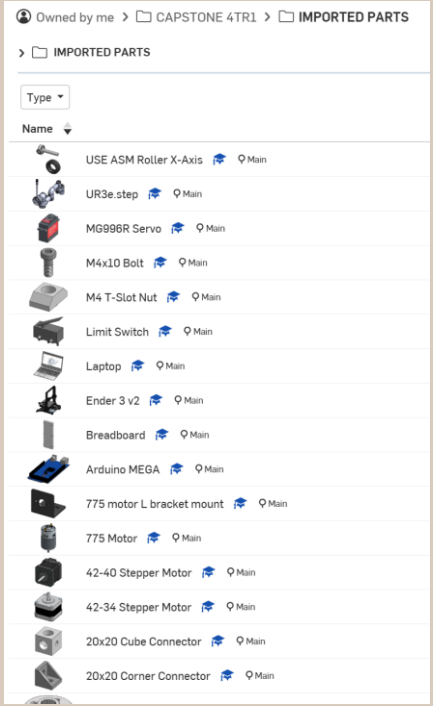
PROGRESS ON CAD DESIGN

We finished the rice rolling station, ribcage roller and linear slide, and we have designs for the robot gripper and slicing station (not shown in image).

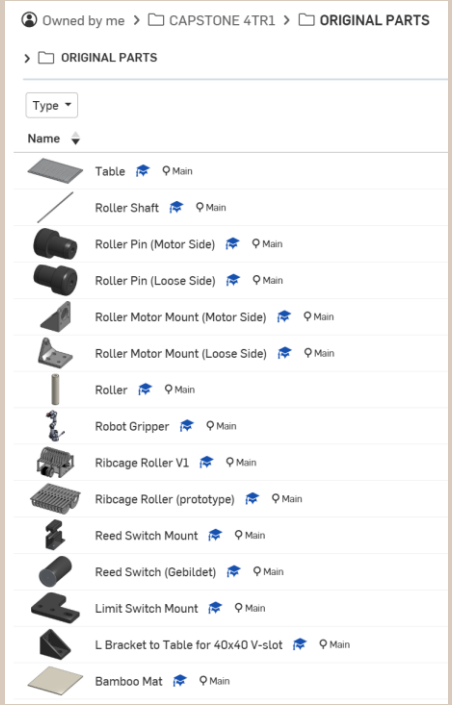
- We have a folder for imported parts and a folder for custom-designed parts
- We have a master assembly file where we import all parts and subassemblies



PROGRESS ON CAD DESIGN



Imported Parts

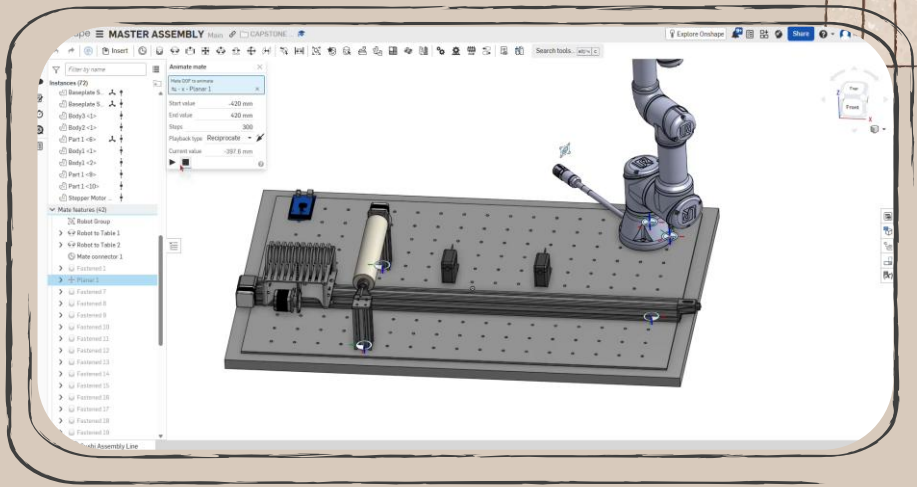


Original Parts

PROGRESS ON CAD DESIGN

Nicolas, Jamie, and Jacob designed the linear slide.

- The slide works and we have tested the control
- Nicolas and Jamie worked on the design and CAD model
- Jacob tested, programmed, and troubleshooted the motors
- All three of us worked on assembly



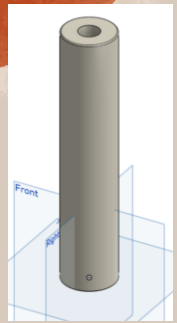
PROGRESS ON CAD DESIGN

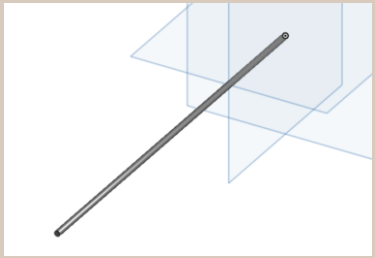
Nicolas and Jamie designed the rice rolling station.

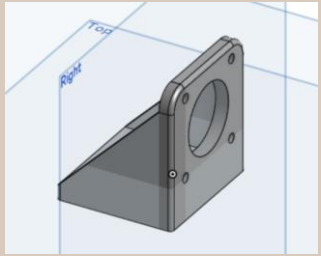
- We kept it simple and had one roller driven by a motor
- The roller is a rolling pin from the dollar store that we disassembled for parts
- Two bolted extrusions that hold the roller at a specific height

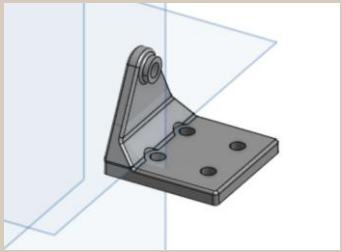


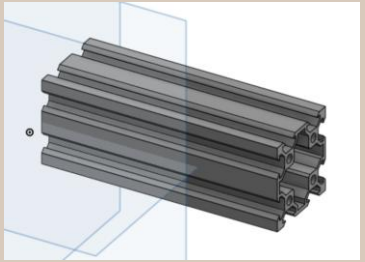
Roller Station Assembly

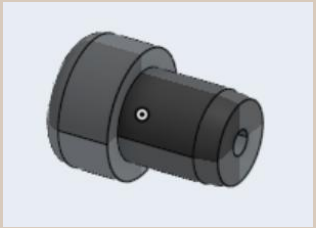


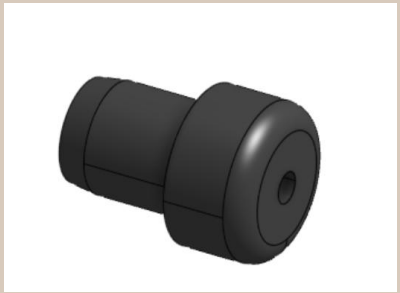


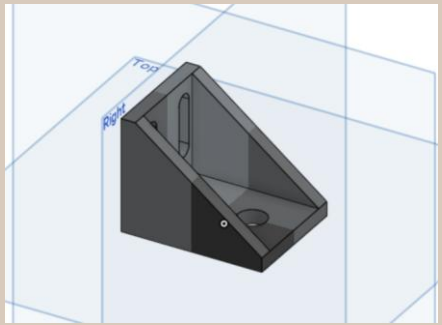


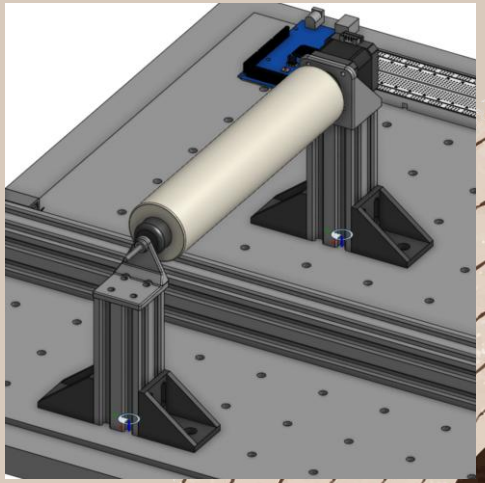








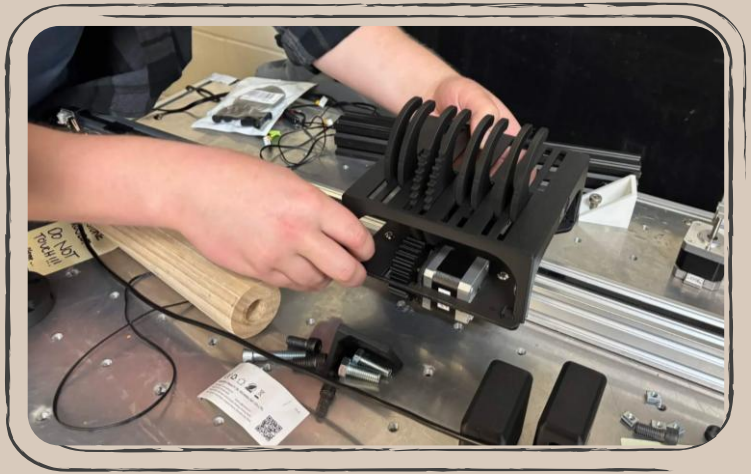




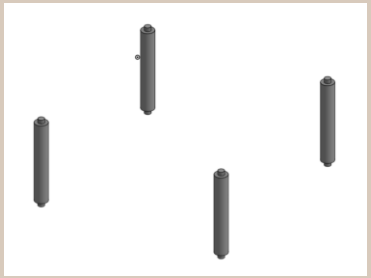
PROGRESS ON CAD DESIGN

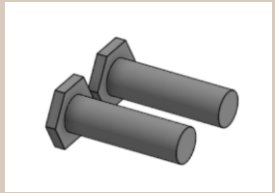
Jamie designed the final design of the ribcage roller.

- He made some major improvements that optimized performance
- He added a mount for the reed switch actuator magnet on the side of the base
- He reduced the number of ribs
- He improved the gear mesh
- He extended the ribs for a tighter roll formation

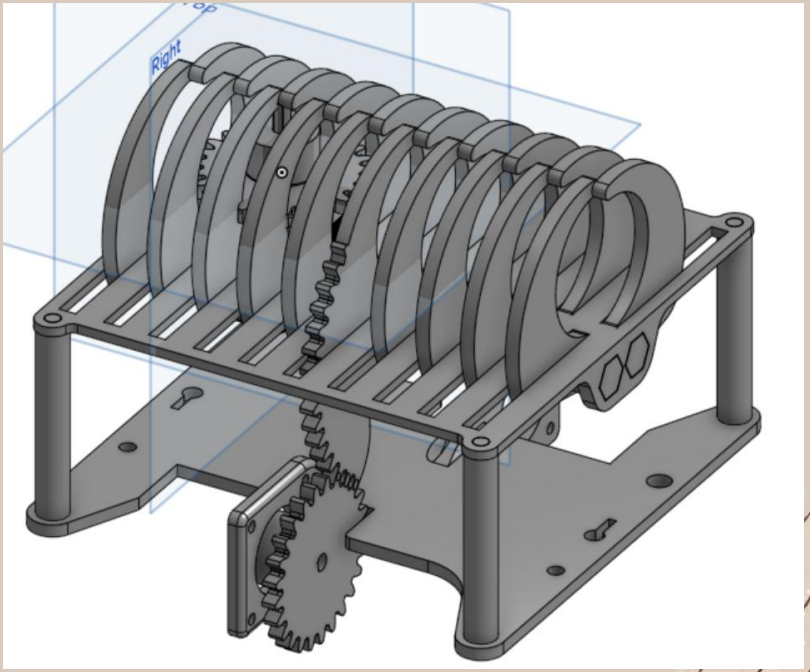


Ribcage Roller Assembly (V1)



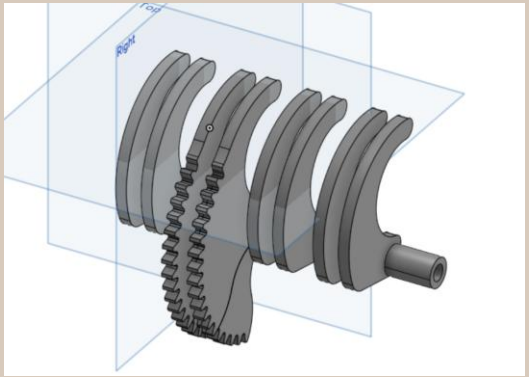




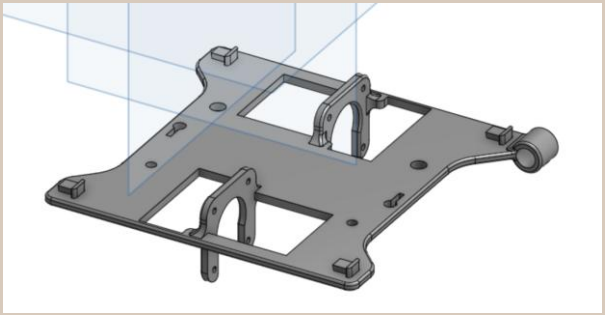


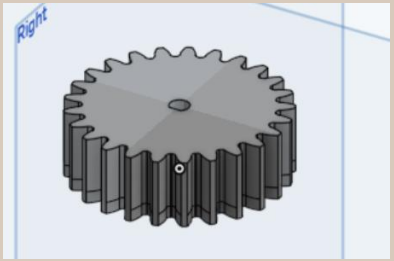
Ribcage Roller Assembly (V2)

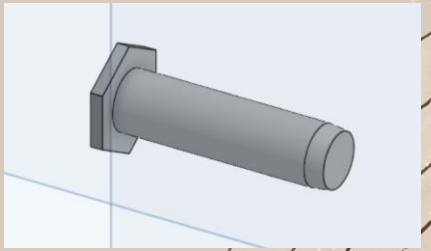








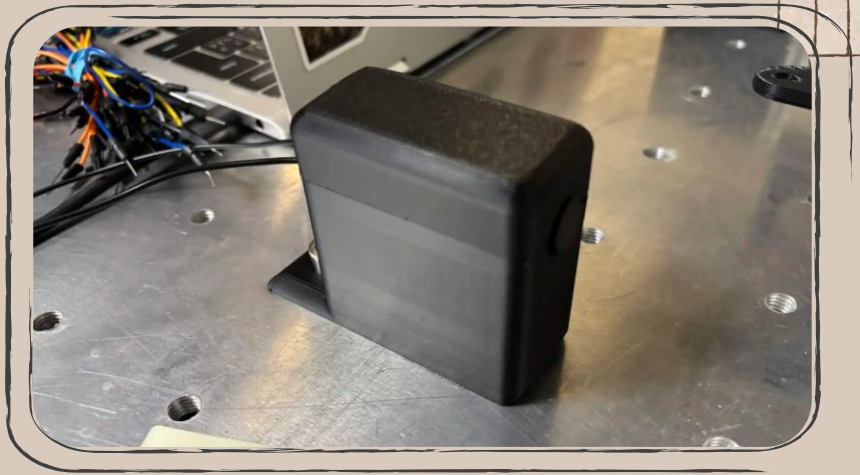




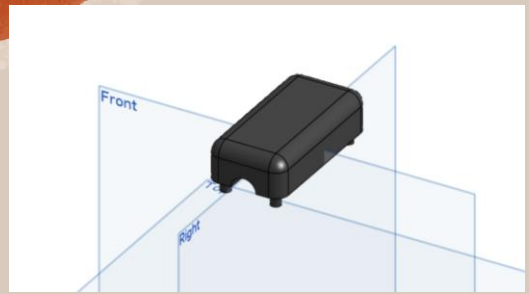
PROGRESS ON CAD DESIGN

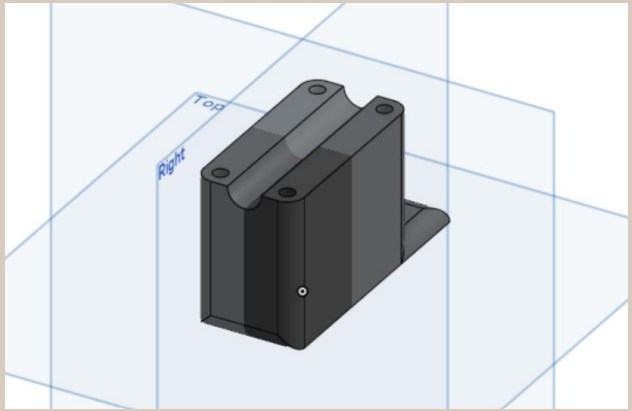
Nicolas designed the reed switch mounts.

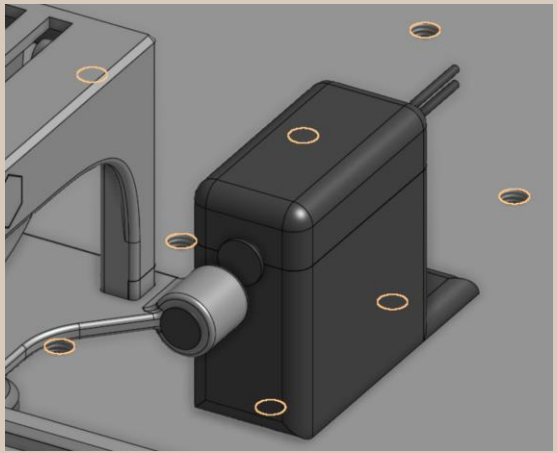
- A detachable cap lid for an easier time to place the sensor inside
- Two holes at the bottom to stabilize it to the table with screws
- At the exact height and distance spec we needed it for the actuator magnet on the platform



Reed Switch Mount Assembly



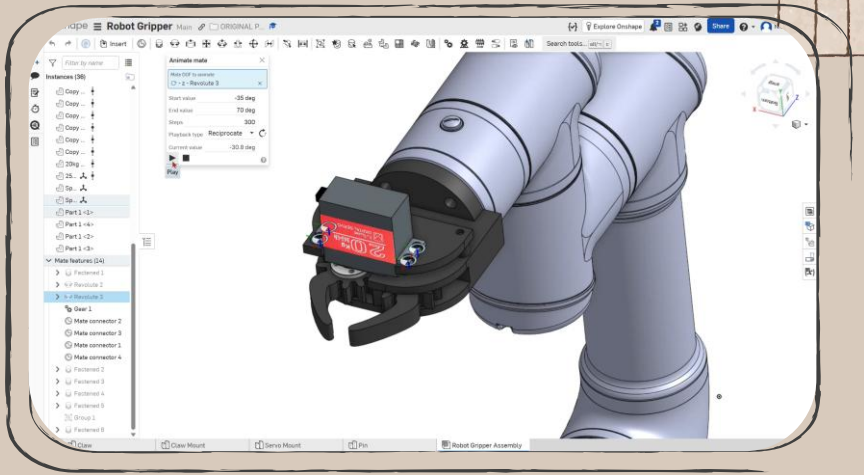




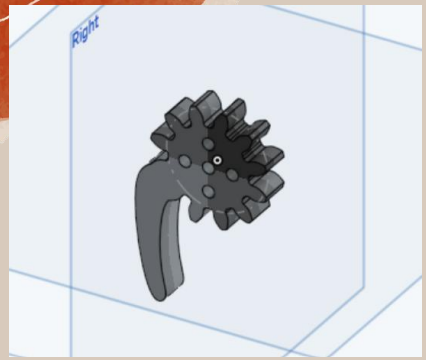
PROGRESS ON CAD DESIGN

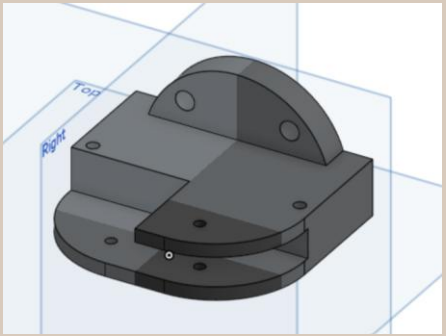
Nicolas designed the robot gripper.

- Just a first draft for now, we need to 3D print it and test it
- Two claws driven by a servo motor
- PWM control and buck converter for digital input 24V to Arduino 5V will be on the table, wires will snake up around the robot arm

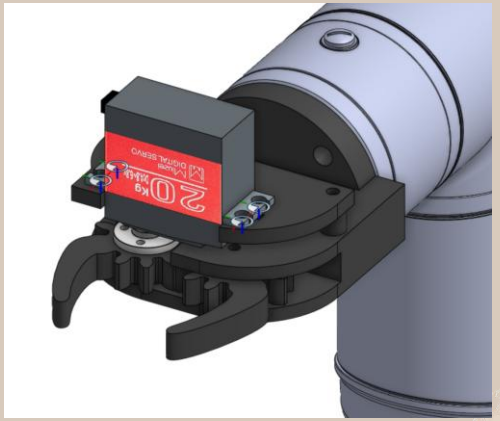


Robot Gripper Assembly



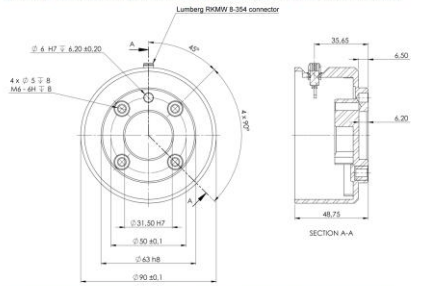






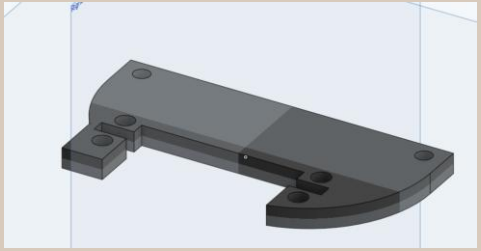
7.11.3. Securing Tool

Description The tool or workpiece is mounted to the tool output flange (ISO) at the tip of the robot.

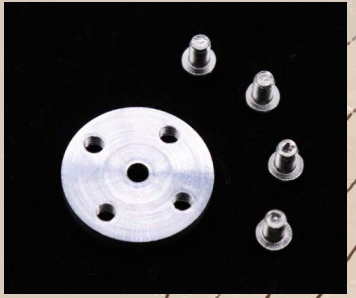


Dimensions and hole pattern of the tool flange. All measurements are in millimeters.

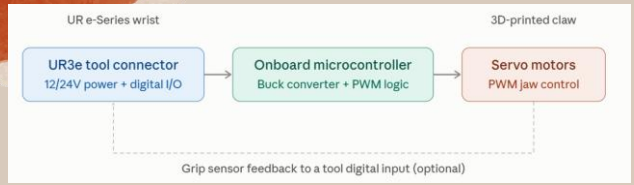
Tool flange The tool output flange (ISO 9409-1) is where the tool is mounted at the tip of the robot. It is recommended to use a radially slotted hole for the positioning pin to avoid over-constraining, while keeping precise position.

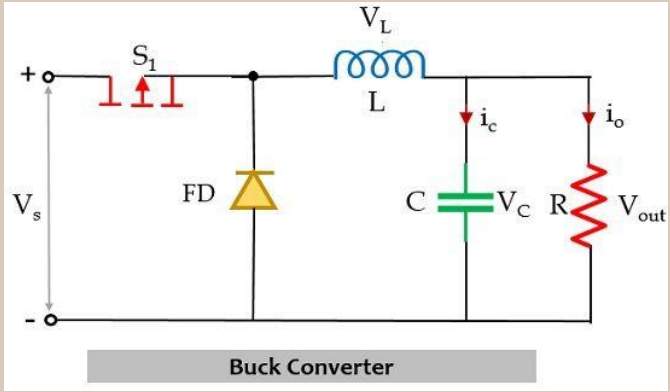






Robot Gripper Control

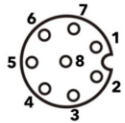




7.11.1. Tool I/O

Tool Connector

The tool connector illustrated below provides power and control signals for the grippers and sensors used on a specific robot tool. The tool connector has eight holes and is located next to the tool flange on Wrist 3.
The eight wires inside the connector have different functions, as listed in the table:



Pin #	Signal	Description
1	AI3 / RS485-	Analog in 3 or RS485-
2	AI2 / RS485+	Analog in 2 or RS485+
3	TO0/PWR	Digital Outputs 0 or 0V/12V/24V
4	TO1/GND	Digital Outputs 1 or Ground
5	POWER	0V/12V/24V
6	TI0	Digital Inputs 0
7	TI1	Digital Inputs 1
8	GND	Ground

NOTICE
The Tool Connector must be manually tightened up to a maximum of 0.4 Nm.

Tool I/O Accessories

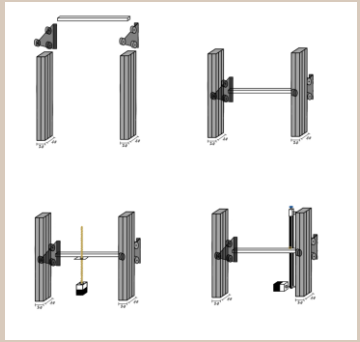
The UR20 tool I/O can require an accessory element to facilitate connection with tools. Depending on the tool, you can use the following tool I/O accessories: Tool Flange Adapter (see Tool Flange Accessories) and/or Tool Cable Adapter.



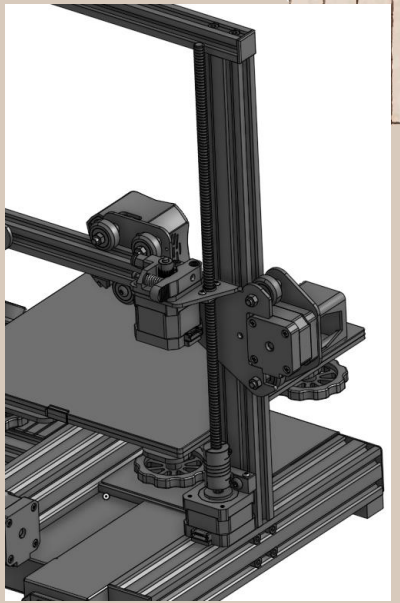
PROGRESS ON CAD DESIGN

Nicolas, Jamie, and Jacob designed the slicing station. Jamie made the CAD model.

- We decided on a lead screw driven by a motor that lowers a frame that holds the blades, which goes in between the ribs







PROGRESS ON CONTROL

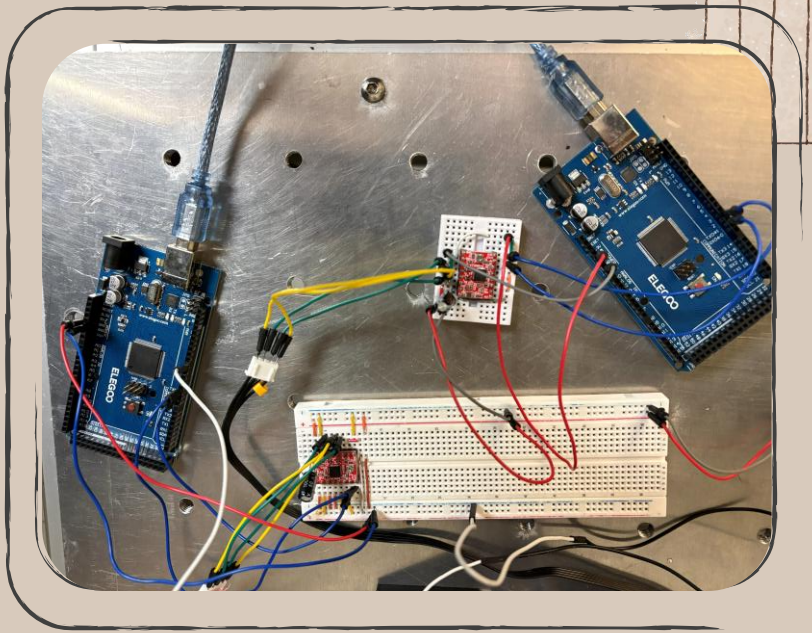
Jacob designed the control system.

- Requires 6 total motors controlled by 2 Arduinos
- Master controller:
  - 2 reed switches
  - 2 limit switches

  Sensors
  - Conveyor belt motor
  - Sushi rolling station motor
  - End-effector motor

  Actuators
  - Robotic arm interface
- Slave controller:
  - I/O connection to Master
  - Rice flattening station motor

  Actuators
  - Slicing station motor



ARDUINO TEST CODE

Reed_Swtich_TestCode | Arduino IDE 2.3.7

File Edit Sketch Tools Help

Arduino Mega or Mega 2...

Reed_Swtich_TestCode.ino

```
void setup() {
  pinMode(stepPin, OUTPUT);
  pinMode(dirPin, OUTPUT);

  pinMode(lsStart, INPUT);
  pinMode(lsEnd, INPUT);

  pinMode(reed1, INPUT_PULLUP);

  Serial.begin(9600);

  digitalWrite(dirPin, HIGH);
}

void loop() {

  bool currentReedState = digitalRead(reed1);

  // Detect a NEW activation of the reed switch
  if (currentReedState == LOW && lastReedState == HIGH) {

    Serial.println("Reed switch detected!");

    // Stop motor for 1 second
    delay(1000);

    Serial.println("Continuing motor...");
  }

  // Update previous state
  lastReedState = currentReedState;

  Serial.println(motorOn);
  // Run motor
  if (motorOn) {
    digitalWrite(stepPin, HIGH);
    delayMicroseconds(800);

    digitalWrite(stepPin, LOW);
    delayMicroseconds(800);
  }
}
```

RiceRoller | Arduino IDE 2.3.7

File Edit Sketch Tools Help

Arduino Mega or Mega 2...

RiceRoller.ino

```
// Stepper motor pins
const int stepPin = 2;
const int dirPin = 3;


bool motorOn = true;

void setup() {
  pinMode(stepPin, OUTPUT);
  pinMode(dirPin, OUTPUT);


  Serial.begin(9600);

  digitalWrite(dirPin, HIGH);
}

void loop() {  //Serial.println(motorOn);
  // Run motor
  if (motorOn) {
    digitalWrite(stepPin, HIGH);
    delayMicroseconds(1200);

    digitalWrite(stepPin, LOW);
    delayMicroseconds(1200);
  }
}
```

NEXT STEPS

1. INSTALLING THE SLICER STATION

2. PROGRAMMING THE ROBOT

3. DEVELOPING THE HMI CONTROL

4. TROUBLESHOOTING & OPTIMIZATIONS



Thanks for listening

First investors get a pack of sushi for free!